

[INTROVERT MASTER PLAN: FINAL BLUEPRINT]

Status: Engineering Phase 100% Complete

[PROJECT VISION]

Introvert is a privacy-centric, serverless, end-to-end communication platform. It utilizes a Hybrid Mesh architecture to ensure resilience and bypass mobile background restrictions while maintaining the highest levels of encryption. The project operates on a sovereign token economy hosted on the Solana network.

- Serverless Mutualism: No central servers; network maintained by users for users.
- Privacy-First: Noise Protocol and Double Ratchet encryption for total sovereignty.
- Sovereign Economy: Circular Introvert Token economy where service provision offsets usage.
- Node Hierarchy:
 - Mobile Clients (Ephemeral): High-efficiency nodes syncing via CRDTs.
 - Desktop Clients (Anchors): Stable backbone nodes providing zero-knowledge “Store & Forward” services.

[TECH STACK & ENVIRONMENT]

- Core Engine: Rust (Systems-level performance and safety).
- Networking: libp2p v0.56.0 (QUIC, WebRTC, Kademlia DHT).
- Identity & Crypto: Ed25519 (Unified Node ID, SQLCipher Key, & Solana Wallet).
- Data Layer: SQLite/SQLCipher with Lamport/Vector Clocks and OR-Set CRDTs.
- Mobile Bridge: Rust C-ABI (FFI) -> Flutter (dart:ffi).
- Blockchain: Solana (Native SPL Token) with a gasless Fee Payer model.

[MODULAR EXECUTION STATUS]

Phase	Description	Status
Phase 1	Core Networking & Identity	[COMPLETE]
Phase 2	Local Data & State	[COMPLETE]
Phase 3	The Anchor Service	[COMPLETE]
Phase 4	VOIP & FFI Bridge	[COMPLETE]
Phase 5	Unified Flutter UI	[COMPLETE]
Phase 6	Solana Incentive Layer	[COMPLETE]
Phase 8	First Light Validation	[COMPLETE]

[SYSTEM ARCHITECTURE SUMMARY]

Layer	Implementation Detail
Identity	Unified Ed25519 identity for Node ID, DB encryption, and Solana Wallet.
Mesh Networking	Resilient libp2p mesh utilizing QUIC/WebRTC for NAT traversal.
Storage	Decentralized SQLCipher with CRDT-based consistency and Vector Clock ordering.
Anchors	Stable nodes providing zero-knowledge mailbox services in exchange for rewards.
Economy	Sovereign Introvert Token model with gasless transactions co-signed by Treasury.

[SYSTEM VALIDATION CHECKLIST]

- ☒ Security: Identity isolated (0600) and derivations salted via HKDF.
- ☒ Mutualism: Reward queuing is non-blocking and integrated into the swarm loop.
- ☒ Causal Integrity: Visualizer renders curved “Chain of Trust” lines based on Vector Clocks.
- ☒ Gasless Logic: IntrovertTokenService isolates users from SOL fee complexity.
- ☒ FFI Reliability: Zero-copy string and pointer handling verified via first_light_integration_test.dart.

[PHASE 8: “FIRST LIGHT” VALIDATION RESULTS]

- Test Environment: Linux host + Flutter FFI Runner.
- Initialization: Rust core successfully derived PeerId and Wallet from secret key.
- FFI Throughput: JSON-serialized identity info correctly passed from Rust to Dart.
- Build Lock: Shared library (libintrovert.so) is verified for desktop and prepared for mobile NDK builds.

The Introvert platform is officially locked and ready for deployment.